Sliding Window Mex

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n integers. Your task is to calculate the mex of each window of k elements, from left to right.

The mex is the smallest nonnegative integer that does not appear in the array. For example, the mex for $[3, 1, 4, 3, 0, 5]$ is 2.

Input

The first line contains two integers n and k: the number of elements and the size of the window.

Then there are n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print $n-k+1$ values: the mex values.

Constraints

- $1 \le k \le n \le 2 \cdot 10^5$
- $0 \le x_i \le 10^9$

Example

Input	Output
`8 3` `1 2 1 0 5 1 1 0`	`0 3 2 2 0 2`